I/O Techniques

Week 8 — Interrupts, Polling, Synchronous/Asynchronous Buses, DMA, and IOP

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Where We Left Off (Week 7)

Last week: how the CPU controls *itself*

- Hardwired control (Week 6) and microprogrammed control (Week 7) both answered the same question: how does the CPU generate its own control-step sequence, cycle by cycle?
- Both designs assumed the CPU is the only thing that matters — registers, ALU, and an internal bus.

The gap Week 7 flagged

I/O devices — keyboards, disks, printers — are far slower than the CPU. Naively waiting for one wastes thousands of cycles. This week: how does the CPU decide what *outside* devices do, and when?

Roadmap: Four Questions, One Thread

1. **Program-controlled I/O (polling)** — what does the CPU do if it simply asks a device "are you ready?" in a loop?
2. **Interrupt-driven I/O** — what if the device asks the CPU instead?
3. **Synchronous vs. asynchronous buses** — one level down: how does the wire-level handshake between master and slave actually work?
4. **DMA and I/O processors** — what if the CPU is removed from the data path entirely?

Running thread for today

A single keyboard-input / display-output pair (`KIN`/`DOUT` status flags) threads through polling and interrupts; a 4096-byte block transfer threads through synchronous/asynchronous timing and into the DMA/IOP Act, where a 3-device arbitration example appears.

Today: who watches the device, and what happens when it's ready?

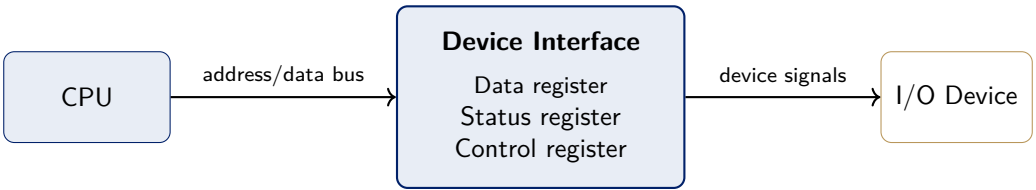
Socratic Question: The Speed Mismatch

Question

A CPU executes an instruction roughly every nanosecond. A keyboard produces one keystroke roughly every *hundred million* nanoseconds. If the CPU must read every keystroke, what should it do in between?

The simplest possible answer: give every I/O device a small set of registers the CPU can read and write *exactly like memory*. This is **memory-mapped I/O** — a device's data/status/control registers occupy addresses in the normal address space, no special I/O instructions needed [1].

The I/O Device Interface



Every interface supplies the same six functions: data/status/control registers, address decoding, timing generation, and format conversion — [1].

Definition: Program-Controlled I/O (Polling)

Polling

In **program-controlled I/O**, the CPU repeatedly reads a device's *status register* in a loop, checking a ready flag, until the device signals it is ready — then performs the data transfer itself.

- New notation: `KIN` — keyboard-input status flag, set when a keystroke is ready in `KBD_DATA`; `DOUT` — display-output status flag, set when the display is ready for the next character in `DISP_DATA`.
- The CPU is the only thing that ever asks; the device never initiates anything.

Worked Example: Polling Loops for KIN and DOUT

Keyboard input

READWAIT loop

```
READWAIT: Test KIN
          Branch=0 READWAIT
          MoveByte KBD_DATA, R
```

Display output

WRITEWAIT loop

```
WRITEWAIT: Test DOUT
           Branch=0 WRITEWAIT
           MoveByte R, DISP_DATA
```

Both loops re-execute the same three instructions on every pass — the CPU is spinning, doing no other useful work [1].

Socratic Check: What's Wasteful About Polling?

Question

A typist produces roughly 5 keystrokes per second. Between two keystrokes, how many `READWAIT` iterations does a 2 GHz CPU execute, and what is it accomplishing on each one?

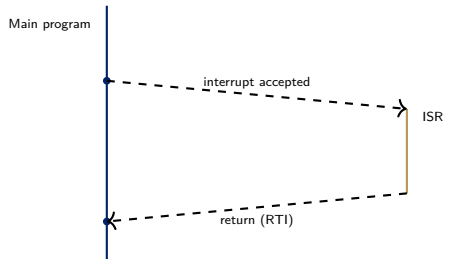
- On the order of hundreds of millions of iterations — every one of them re-reading a status flag that has not changed, accomplishing *nothing* useful.
- Polling is simple and requires no extra hardware, but it wastes CPU time in proportion to how much slower the device is than the CPU. Fine for a single slow device with nothing else to do; wrong at scale.

What if the device asked the CPU, instead of the CPU asking the device?

Definition: Interrupt

Interrupt

An **interrupt** is a signal from a device that suspends the CPU's current program, transfers control to a fixed handler — the **interrupt-service routine (ISR)** — and resumes the original program afterward, exactly where it left off.



The gap between the request and the jump is **interrupt latency**; registers the ISR must not disturb are saved in **shadow registers** [1].

Enabling and Disabling Interrupts

Two independent switches

- `IE` — a single interrupt-enable bit inside the processor status register (`PS`); when `IE`$= 0$ the CPU ignores *every* interrupt request, regardless of source.
- A per-device interrupt-enable bit in that device's control register (`KIRQ` for the keyboard, `DIRQ` for the display) — controls whether *this* device is allowed to request an interrupt at all.

A request reaches the CPU only when both switches agree: the device's own enable bit *and* `PS.IE` [1].

Worked Example: A 5-Step Interrupt-Handling Scenario

1. Device sets its status flag (e.g. `KIN`$= 1$) and, if `KIRQ`$= 1$, raises an interrupt-request line.
2. CPU finishes its *current instruction* (never mid-instruction), then checks `PS.IE`.
3. If enabled: CPU saves `PC` (and any registers the ISR will clobber) into shadow registers, disables further interrupts, and loads `PC` with the ISR's starting address.
4. ISR services the device (e.g. reads `KBD_DATA`, clearing `KIN`), then executes a return-from-interrupt instruction.
5. Return-from-interrupt restores the saved `PC` and re-enables interrupts — the interrupted program resumes with no memory of having been interrupted [1].

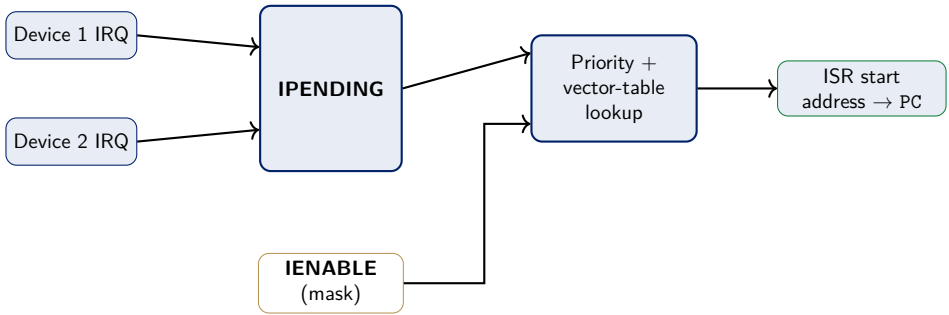
Handling Multiple Devices

Two designs for "which device interrupted?"

- **Polling the sources**: on any interrupt, the ISR checks each device's status flag in turn — simple, but the fixed check order costs time proportional to the number of devices.
- **Vectored interrupts**: each device supplies its own identifying code; hardware uses it to index directly into an **interrupt-vector table** — a table of ISR starting addresses, one entry per device — and jumps straight there.

Vectored interrupts trade a small amount of hardware (the vector table and the lookup) for constant-time dispatch, regardless of device count [1].

Interrupt Nesting and Priority



When both devices request simultaneously, the priority encoder grants the higher-priority one first; `IPENDING` keeps the other's request latched until it is serviced [1].

Socratic Check: Simultaneous Interrupt Requests

Question

Device 1 (higher priority) and Device 2 both request an interrupt on the same cycle. Device 1's ISR is still running when Device 2's request would normally be granted. What happens?

- If **nesting** is allowed and Device 2 outranks whatever priority level Device 1's ISR is running at, Device 2 can interrupt *that ISR* — the same 5-step scenario applies recursively.
- If nesting is disabled (or Device 2 is lower priority), Device 2's request stays latched in `IPENDING` until Device 1's ISR re-enables interrupts.

Polling vs. Interrupts: Trade-offs

	Polling	**Interrupts**
Who initiates	CPU (asks repeatedly)	Device (signals once, when ready)
CPU cost while waiting	Wasted spin cycles	Zero — CPU runs other code
Extra hardware	None	`IE`, per-device enables, vector table
Response latency	Depends on poll rate	Interrupt latency (usually smaller)
Best fit	One slow device, nothing else to do	Multiple/unpredictable devices

Same problem, opposite direction of initiative — interrupts move the waiting cost from the CPU to the hardware that detects readiness.

One level down: what does “the device is ready” look like on the wires?

Bus Structure: Master and Slave Roles

One shared bus, many devices

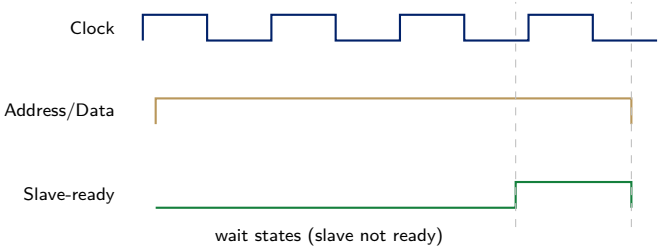
Every I/O interface in Act 1's diagram in fact hangs off a single shared **bus** — address, data, and control lines every device listens to. A **bus protocol** defines how a **master** (the device driving the transfer, usually the CPU) and a **slave** (the responding device) coordinate one data transfer without colliding.

This Act asks: when a master says "take this data" or "give me that data," how does the slave say "I'm not ready yet" or "here it is"? Two different answers: **synchronous** and **asynchronous** [1].

Definition: Synchronous Bus

Synchronous bus

All devices share a common **bus clock**. The master places an address/data on the bus at a fixed clock edge; every slave samples the bus at the *same* edge, with no per-transfer negotiation.

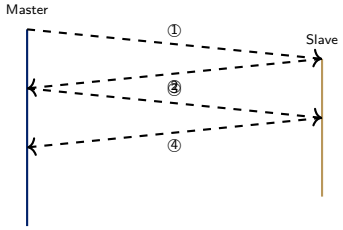


A slow slave stretches the transfer by holding `Slave-ready` low for extra clock cycles — the master simply waits, still synchronized to the shared clock [1].

Definition: Asynchronous Bus (Full Handshake)

Asynchronous bus

No shared clock. Master and slave exchange explicit **Master-ready**/**Slave-ready** signals, each one only changing *in response to* the other — a **full handshake**.



① Master-ready asserted
② Slave-ready (data valid)
③ Master-ready withdrawn (latched)
④ Slave-ready withdrawn

Each step waits for the previous one — **bus skew** (wires arriving at slightly different times) never causes a false sample, unlike a synchronous bus's single fixed clock edge [1].

Synchronous vs. Asynchronous: Trade-offs

	Synchronous	**Asynchronous**
Timing reference	Shared bus clock	Explicit handshake, no clock
Slow-device handling	Extra wait-state clock cycles	Handshake simply takes longer
Skew sensitivity	Sensitive — must beat clock period	Immune — self-timed
Per-transfer overhead	Low (one clock edge)	Higher (two round trips)
Best fit	Devices of similar, known speed	Mixed-speed / unpredictable devices

Same coordination problem, opposite mechanism — a shared clock everyone trusts, or an explicit conversation that trusts nothing.

Socratic Check: Why Not Always Use Asynchronous?

Question

Asynchronous handshaking tolerates skew and adapts automatically to any device speed. Why do most high-speed processor buses use synchronous protocols instead?

- Every asynchronous transfer pays *two full round-trip delays* (steps 1–2 and 3–4) — for devices that are consistently fast, this is pure overhead a synchronous bus does not pay.
- Synchronous buses only lose when a device is unusually slow (extra wait states) or the clock period must shrink to accommodate skew — exactly the trade-off the comparison table names.

Interrupts still cost one instruction per word transferred — what if the CPU left the loop entirely?

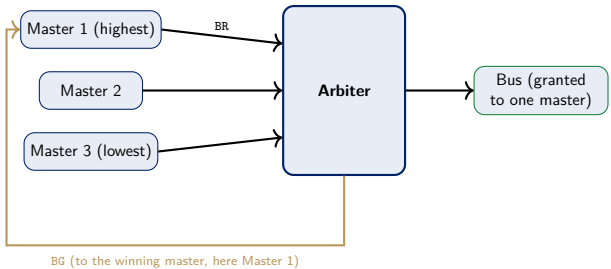
The Cost Interrupts Still Pay

Socratic question

A disk transfer moves a 4096-byte block, one 4-byte word at a time. With interrupt-driven I/O, the ISR executes on *every word*: read status, read/write data, update a memory pointer, decrement a counter, return. What does this cost at 4096 bytes, one interrupt per word?

$4096/4 = 1024$ ISR invocations for a *single* block transfer — interrupts removed the CPU's *waiting*, but not its involvement in *moving* the data. The next step removes the CPU from the data path itself.

Bus Arbitration: A Third Role for a Device



Any device connected via BR/BG can become bus **master**, not just the CPU — the arbiter grants the bus to exactly one requester, by priority, each time [1].

From Arbitration to Direct Memory Access

The functional idea, stated plainly

Once a device can become bus master, it can transfer data *directly to or from memory* — no CPU instruction touches the data at all. The CPU is interrupted *once*, when the whole block finishes, not once per word.

This edition of Hamacher's text (Ch. 7, arbitration) describes exactly this mechanism but never names it "DMA" — the term and its standard treatment below follow Stallings Ch. 8 [2], per the syllabus and `.claude/rules/textbook-grounding.md`.

Definition: Direct Memory Access (DMA)

DMA (general/standard treatment — Stallings Ch. 8)

A **DMA controller** is given a source/destination address and a word count by the CPU, then becomes bus master (via arbitration) and transfers the entire block directly to/from memory, interrupting the CPU only when the *block* — not each word — is complete [2].

- Pro: CPU involvement drops from "one ISR per word" — 1024 of them for our 4096-byte block — to "one setup, one completion interrupt" — a constant cost regardless of block size.
- Con: DMA controller and CPU now compete for the same memory bus (**cycle stealing**); extra hardware required.

Definition: I/O Processor (IOP)

IOP (general/standard treatment — Stallings Ch. 8)

An **I/O processor** generalizes DMA one step further: a dedicated processor, given a whole *channel program* (a short program describing several transfers, formats, and simple decisions), executes it independently — interrupting the main CPU only when the entire program completes, not per block.

Where DMA offloads *moving* data, an IOP offloads *managing* a sequence of I/O operations — the difference is how much decision-making is delegated away from the CPU, not just how the data moves.

Four I/O Techniques, Head to Head

	Polling	**Interrupts**	**DMA**	**IOP**
CPU cost / word	Every poll	Every word (ISR)	None (setup only)	None
CPU cost / block	—	Thousands of ISRs	One interrupt	One interrupt
Extra hardware	None	Vector table, `IE`	DMA controller	Dedicated processor
Best fit	One slow device	Multiple/sporadic devices	Bulk block transfer	Complex I/O subsystems

Same core question, four answers: who watches the device, and how much of the CPU does watching cost?

Socratic Check: Choosing a Technique

Question

A system reads one temperature sensor value per second (low rate, latency-tolerant) and separately copies a 10 MB file from disk to memory (high rate, throughput-critical). Which technique fits each?

- Temperature sensor: **interrupts** — rate is low, so ISR overhead per event is negligible, and the CPU is free the rest of the time.
- Disk copy: **DMA** — per-word CPU involvement at that data rate would dominate CPU time; DMA reduces it to one setup, one completion interrupt.

The Four Threads Meet

Every design choice moves the waiting cost somewhere else

- **Polling vs. interrupts (programmer's view):** polling wastes CPU cycles asking; interrupts let the device ask instead, at the cost of vector-table/priority hardware.
- **Synchronous vs. asynchronous (bus hardware):** a shared clock is cheap per transfer but skew-sensitive; a handshake is skew-immune but pays two round trips every time.
- **Interrupts vs. DMA/IOP (who moves the data):** interrupts still touch every word; DMA and IOP remove the CPU from the data path, trading extra hardware for near-zero per-word cost.

None of these is free — exactly the pattern Weeks 5–7 established for buses and control units, now applied to the world outside the chip.

Bridge to Week 9

From moving data to storing it efficiently

Weeks 1–8 built a CPU that computes, controls itself, and now moves data to and from slow devices. Week 9 asks: once data is in memory, how do we make *memory itself* fast enough to keep up with the CPU?

- DRAM is far slower than the CPU — the same speed-mismatch problem this week solved for devices reappears for memory itself.
- Cache memory, mapping functions, and replacement policies are Week 9's answer.

Summary

- **Memory-mapped I/O**: device registers (data/status/control) live in the normal address space; **program-controlled I/O (polling)** repeatedly tests a status flag (`KIN`, `DOUT`).
- **Interrupts**: the device signals readiness; `PS.IE` and per-device enables (`KIRQ`/`DIRQ`) gate delivery; **vectored interrupts** use `IPENDING`/`IENABLE` plus an interrupt-vector table for constant-time dispatch.
- **Synchronous bus**: shared clock, wait states for slow slaves. **Asynchronous bus**: `Master-ready`/`Slave-ready` full handshake, skew-immune but higher per-transfer overhead.
- **Bus arbitration** (`BR`/`BG`) lets a device become bus master — the functional basis for **DMA** (block transfer, one completion interrupt) and **IOP** (whole channel programs, general/standard treatment per Stallings Ch. 8).
- **Four techniques, one trade-off axis**: how much of the CPU does watching and moving the data cost, and how much hardware buys that cost down.

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